

IEM KOLKATA
Linear Algebra
Eigenvalues & Eigenvectors

Q1) If A is a diagonal matrix, which of the following is true?

- (a) The eigenvalues of A are the entries on its diagonal.
- (b) A is invertible if and only if all its diagonal entries are non-zero
- (c) A is symmetric if all its diagonal entries are real.
- (d) All of the above.

Q2) Let $1, -1, 0$ be eigenvalues of matrix A , then $|A^{100} + I| = ?$

Q3) Consider the matrix as given below.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 7 \\ 0 & 0 & 3 \end{bmatrix}$$

Which one of the following options provides the CORRECT values of the eigenvalues of the matrix?

- (a) 1, 4, 3
- (b) 3, 7, 3
- (c) 7, 3, 2
- (d) 1, 2, 3

Q4) Suppose that the eigenvalues of matrix A are 1, 2, 4. The determinant of $(A^{-1})^T$ is _____

- (a) $1/8$
- (b) 1
- (c) $1/4$
- (d) 2

Q5) At least one eigenvalue of a singular matrix is

(a) Positive (b) Zero (c) Negative (d) Imaginary

Q6) Consider the following matrix

$$\begin{bmatrix} 2 & 3 \\ x & y \end{bmatrix}$$

If the eigenvalues of A are 4 and 8, then

(a) $x=-3$, $y=9$

(b) $x=-4$, $y=10$

(c) $x=5$, $y=8$

(d) $x=4$, $y=10$

Q7) The product of all eigenvalues of the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

is

(a) -1 (b) 0 (c) 1 (d) 2

Q8) Let A be the 2×2 matrix with elements $a_{11} = a_{12} = a_{21} = +1$ and $a_{22} = -1$. Then the eigenvalues of the matrix A^{19} are

(a) 1024 and -1024

(b) $1024\sqrt{2}$ and $-1024\sqrt{2}$

(c) $4\sqrt{2}$ and $-4\sqrt{2}$

(d) $512\sqrt{2}$ and $-512\sqrt{2}$

Q9) The matrix $\begin{bmatrix} 1 & a \\ 8 & 3 \end{bmatrix}$ (where $a > 0$) has a negative eigenvalue if a is greater than

(a) $\frac{3}{8}$ (b) $\frac{1}{8}$ (c) $\frac{1}{4}$ (d) $\frac{1}{5}$

Q10) Let A be a 2×2 matrix as given

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

What are the eigenvalues of the matrix A^{13} ?

(a) 1, -1 (b) $2\sqrt{2}$, $-2\sqrt{2}$ (c) $4\sqrt{2}$, $-4\sqrt{2}$ (d) $64\sqrt{2}$, $-64\sqrt{2}$

Q11) Let

$$P = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 1 & 1 \\ -1 & -1 & 0 & 1 & 1 \\ -1 & -1 & -1 & 0 & 1 \\ -1 & -1 & -1 & -1 & 0 \end{pmatrix}$$

If $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ and λ_5 are eigen values of P , then $\prod_{i=1}^5 \lambda_i$ equals _____
(answer in integer).

Q12) Let $P = (a_{ij})$ be a 10×10 matrix with

$$a_{ij} = \begin{cases} -\frac{1}{10} & \text{if } i \neq j \\ \frac{9}{10} & \text{if } i = j \end{cases}$$

Then $\text{rank}(P)$ equals

(A) 10 (B) 9 (C) 1 (D) 8

Q13) Let $P = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $Q = \begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix}$ Then the value of $\text{trace}(P^5 + Q^4)$ equals _____ (answer in integer).

Q14) Let $P = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$ and $Q = P^3 - 2P^2 - 4P + 13I_2$, where I_2 denotes the identity matrix of order 2. Then the determinant of Q is equal to _____ (answer in integer).